

Arbitraging Covered Interest Rate Parity Deviations and Bank Lending[†]

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I propose and test a new channel through which covered interest rate parity (CIP) deviations can affect bank lending in emerging economies. I argue that when CIP deviations exist, banks attempt to arbitrage them. To do so, banks must borrow in a particular currency. When this currency is scarce, bank lending in the currency required to arbitrage decreases, while they use this currency in their arbitrage activities. I test this channel by exploiting differences in the abilities of Peruvian banks to arbitrage CIP deviations. I find evidence that supports the proposed channel. (JEL F31, G01, G13, G21, O16)

The covered interest rate parity (CIP) condition is the fundamental pricing equation for foreign exchange rate forward and swap contracts. Deviations from CIP occur in developing and developed nations alike, but for different reasons and with a different dynamic. Research has focused mostly on understanding how these deviations from CIP affect asset prices or in which environments they arise. However, not much is known about possible channels through which these deviations could end up affecting households and firms.

In this paper, I argue that CIP deviations could have broader effects than previously known. I show that, in an emerging-market setting, when banks arbitrage part of these deviations, they change the currency composition of their lending. This is important for the real economy. The literature has extensively demonstrated that firms' and households' currency mismatches, led by the currency composition of their debt, have serious effects on economic activity after a local currency depreciation. This paper complements the existing research on currency mismatches by

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suggesting that CIP deviations can have real effects and showing that CIP deviations can impact these mismatches through a currency recomposition of banks' lending.

I start by proposing a channel through which CIP deviations can affect bank lending in a partially dollarized economy. The channel works as follows: CIP deviations imply that arbitrage opportunities exist. Banks, as the natural arbitrageurs, attempt to arbitrage them. However, the arbitrage requires banks to borrow a particular currency. When this currency is scarce, banks must either increase deposit rates or reduce the currency's use in other activities, such as lending, in order to use these funds to arbitrage. Consequently, lending in the currency needed to arbitrage will likely decrease relative to the one that is not required. Hence, arbitraging CIP deviations can contribute to changes in currency mismatches in partially dollarized economies.

I test this channel by studying the relationship between arbitraging CIP deviations and bank lending in Peru during an eight-year period from 2005 to 2013, excluding the global financial crisis (GFC).¹ I proceed in three steps. First, I demonstrate that banks' exchange rate and money market transactions suggest that they arbitrage CIP deviations. Second, I show that banks' funding in the currency required to arbitrage CIP deviations becomes scarcer or more expensive as CIP deviations increase. Third, I exploit banks' varying arbitrage sensitivities to CIP deviations to show that banks with higher arbitrage activity shift the currency in which they lend by more, in line with how the arbitrage is done.

How is CIP arbitrage executed? Consider a local bank in Peru with the opportunity to lend for one month at the risk-free rate in either dollars or in *soles*, the Peruvian currency. Under CIP, the return on lending soles should equal the return on lending dollars, while simultaneously hedging the exchange rate risk by selling dollars forward to convert them back to soles. The return of the combination of lending dollars and hedging the exchange rate is the *soles synthetic rate*. The soles synthetic rate minus the soles cash rate is the *cross-currency basis*. The cross-currency basis measures the deviations from CIP.

Based on this description of how CIP is done, step one of my empirical analysis uses confidential data from Peru to show that banks' transactions are consistent with arbitraging CIP. The data include all of the forward contracts of all banks in Peru and all of their daily spot transactions, as well as the banks' interbank loans, financial obligations, and investments. With this data, I show that consistent with the arbitrage, when the cross-currency basis increases and the soles synthetic rate is greater than the soles cash rate, banks borrow at the lower soles cash rate and lend at the higher soles synthetic rate.

Arbitraging CIP deviations should not affect other markets, such as the credit market, if banks could easily fund the currencies required to do the arbitrage. But if the currencies banks need to fund the arbitrage are scarce when carrying out the arbitrage, there can be a pass-through to the lending market.

Hence, step two of my empirical analysis shows that the currency banks need to borrow for the arbitrage is scarce. To show the scarcity, I complement the analysis with confidential information about daily bank-level interest rates paid on bank

¹ While I end the sample in 2013 due to regulatory confounders in Peru, my results hold until the end date of my datasets in 2018.

deposits. With this data, I find that as the cross-currency basis increases, banks only pay higher deposit spreads and have lower liquidity on the currency that is required to arbitrage.

Therefore, given that banks seem to arbitrage CIP deviations but funding the currency required for the arbitrage is scarce, step three of my empirical analysis studies whether there are consequences in the lending market of banks doing the arbitrage in such conditions.

Step three relies on an important finding: while banks' transactions are consistent with arbitraging CIP deviations, banks differ significantly in their response to these CIP deviations.

The finding that banks differ in arbitrage sensitivity derives from two computations. First, I compute a proxy for each bank's arbitrage position, which is the amount of a bank's forward position that is matched with a spot position in opposite direction. I use this proxy because every arbitrage transaction requires banks to offset their forward position with spot positions of similar magnitude. To prevent the results from being affected by banks' size, I scale each bank's arbitrage position by their assets. I call the resulting variable "*Matched/Assets*." Second, for each bank, I regress the banks' "*Matched/Assets*" variable on the market cross-currency basis. The resulting coefficient, $\hat{\beta}_b$, is the arbitrage sensitivity—the amount banks change their arbitrage positions after a change in the cross-currency basis. I find significant differences across banks' arbitrage sensitivities, which seems associated with the type of customer banks have in the forward market.

To analyze the impact of arbitraging CIP deviations on bank lending in soles and dollars, I use (i) the bank-specific measure of banks' sensitivities to arbitrage CIP deviations, $\hat{\beta}$; and (ii) the Chilean and Mexican cross-currency basis as an instrument for Peru's cross-currency basis (USDPEN). The heterogeneity in banks' arbitrage sensitivities to CIP deviations, coupled with the fact that 70 percent of firms in my sample borrow from multiple banks, allows me to use a within-firm-month analysis. This approach enables me to control for shocks that affect all banks and for firms' credit demand. Using the instrument for Peru's basis helps reduce the influence of shocks to the Peruvian economy that correlate with CIP deviations in Peru and bank lending. It also helps mitigate concerns that the particular hedging choices of banks affect the cross-currency basis and, hence, the effects of such hedging on my results.

Making use of banks' differences in arbitrage sensitivities and of the instrument, I show that banks that allocate 1 percentage point more of their assets to arbitraging CIP deviations increase their dollar lending relative to soles lending between 11 percent and 41 percent after a 1 percentage point increase in the USDPEN cross-currency basis instrument.² This increase in the difference between dollars and soles lending is due to both an increase in dollars lending and a decrease in soles lending. Since total lending is barely affected, in net, the results show a recomposition of banks' balance sheets rather than a change in size.

While the within-firm-month analysis allows me to control for shocks that affect all banks, banks are heterogeneous, and therefore, shocks might not affect them to

²This is, $\log(USD) - \log(PEN)$. The most conservative estimate I obtained is 11 percent, which holds when restricting the sample to firms borrowing in soles and dollars. When including all firms and the possibility of firms switching currencies, the estimates increase to 41 percent.

the same degree. To address this concern, I use lagged bank controls and perform robustness checks that restrict the sample to very similar banks. The results persist.

My results are also robust to various other concerns. I provide robustness checks regarding correlations between macroeconomic factors and CIP deviations, such as that between the exchange rate and CIP deviations (Avdjiev et al. 2019). In online Appendix C, I also show that the institutional setting and exchange rate-related policies do not affect my results. Additionally, my results are robust to a series of alternative specifications, including using different measures of CIP deviations, different samples, firms, and exposure measures to sort banks by arbitrage intensity.

To the best of my knowledge, this is the first paper to propose and empirically test a channel through which, by arbitraging CIP deviations, banks can change the currency composition of their lending portfolios. In this channel, I show that when banks arbitrage, they face liquidity constraints in the currency they leverage to arbitrage the deviations. Consequently, when engaging in the arbitrage, they reduce lending of the currency they leverage for arbitrage. Instead, they increase lending in the currency that they would have in excess after the arbitrage.

While I focus on a partially dollarized economy, it is likely that the contributions of this paper extend to other economies. Many countries where local banks do not actively take foreign currency deposits from local households and firms have important shares of foreign currency loans. The IMF's Financial Soundness Indicator lists several countries with more than 30 percent of foreign currency loans, including Canada, Luxembourg, United Kingdom, Singapore, and Switzerland. Given that global banks borrow and lend in various economies, arbitraging CIP deviations could affect the allocation of global banks' currency portfolios (and respective borrowing and lending) across countries. Therefore, the mechanism outlined in this paper could apply on an international scale.

This paper complements three areas of research relative to current literature. First, it complements the current literature on CIP deviations by providing evidence that suggests that CIP deviations affect more than just arbitrage activities; they can also affect firms and households through changes in the currency composition of their loans. This is novel in the literature of CIP deviations, which has mainly addressed the predominance of CIP deviations after the GFC (Baba, Packer, and Nagano 2008; Baba and Packer 2009; Coffey, Hrungr, and Sarkar 2009; Mancini-Griffoli and Ranaldo 2011; Du, Tepper, and Verdelhan 2018)³ as well as addressed why these deviations exist, what can be correlated with them, or environments in which these may appear (Borio et al. 2018; Du, Tepper, and Verdelhan 2018; Avdjiev et al. 2019; Wallen 2022; Correa, Du, and Liao 2020; Puriya and Bräuning 2020; Rime, Schrimpf, and Syrstad 2022; Cenedese, Della Corte, and Wang 2021; Cerutti, Obstfeld, and Zhou 2021; Dedola et al. 2021; Aizenman, Ito, and Pasricha 2022; Du, Hébert, and Huber 2022; Liao and Zhang 2021; Aldunate et al. 2023; De Leo, Gopinath, and Kalemli-Özcan 2023; Cerutti and Zhou forthcoming). Except for a few papers such as Cerutti and Zhou (forthcoming) and De Leo, Gopinath, and Kalemli-Özcan (2023), most papers focus on developed economies. Because

³ Akram, Rime, and Sarno (2008) report that in developed countries, CIP deviations can be seen even before the GFC in high frequency but they are short-lived. However, most research renders these quite small at lower frequencies. In developing economies, these have been large, even before the GFC.

most of my sample exhibits carry trade inflows in Peru, my setting here resembles that in Keller (2020) and Amador et al. (2020). In both cases, CIP deviations arise in a setting with capital inflows and a central bank that intervenes in the exchange rate to mitigate exchange rate appreciation.⁴

Second, my paper complements the literature on the real effects of currency mismatches after depreciation shocks by showing that CIP deviations shift the currency in which firms borrow. In an environment with very limited hedging, which is the case in Peru (and also occurs in other countries (see Alfaro, Calani, and Varela 2023; Levin-Konigsberg et al. 2023), firms and households' currency mismatches are typically caused by changes in the currency composition of debt. The literature shows that these mismatches have important effects on the real economy, such as financial distress, reduced demand, increased unemployment, and worse recessions (Verner and Gyöngyösi 2020). Firms with foreign currency debt also decrease investment and increase exit rates (Aguiar 2005; Kim, Tesar, and Zhang 2015; Hardy 2023; Verner and Gyöngyösi 2020; Du and Schreger 2022). The effects are worse for smaller firms (Hardy 2023) and for domestic-owned firms amid a currency and banking crisis (Kalemli-Özcan, Kamil, and Villegas-Sanchez 2016). However, before the depreciation shock, borrowing in dollars can relax financial constraints (Endrész and Harasztosi 2014; Rancière and Tornell 2016; Verner and Gyöngyösi 2020; Salomao and Varela 2022). Therefore, my paper suggests that CIP deviations could affect the real economy by showing that CIP deviations can affect these mismatches.

I lack firm-level data and a setting that would allow me to identify the effects of the shift in the currency composition of firms' borrowing on their outcomes. However, these effects are likely present in Peru. For example, Humala (2019) finds that in Peru, the 30 percent depreciation of the sol between 2013 and 2015 was associated with earnings losses of between 24 percent and 36 percent. Similarly, a survey by EY Peru⁵ found that profits declined by between 15 percent to 20 percent, and that 75 percent of the firms in their sample were affected by the 10 percent depreciation of the sol in 2021. The Central Bank of Peru has said that the possibility of such negative effects has also shaped its regulations, including the de-dollarization regulation that occurred during the depreciation associated with the taper tantrum in 2013.

Third, my paper contributes to the literature on understanding of internal capital markets in the banking system by presenting new empirical evidence concerning how internal capital markets work for a bank that must allocate scarce currency-specific liquidity between its lending and trading divisions. The empirical evidence has mostly focused on diversified firms (Lamont 1997; Shin and Stulz 1998), bank holding companies (Houston, James, and Marcus 1997; Houston and James 1998; Campello 2002; Ashcraft and Campello 2007; Cremers, Huang, and Sautner 2011), and global banks (Cetorelli and Goldberg 2012a, b; Bruno and Shin 2015; Cao and Dinger 2022). Papers on reallocation of funds within a bank in a single country

⁴Gutierrez, Ivashina, and Salomao (2023) study a different question using Peruvian data. However, their setting differs from mine. Their sample starts in 2012, and they cover a period characterized by a significant depreciation of the Peruvian sol and various regulations in the lending and forward markets that act as significant confounders for studying the effect of arbitraging CIP deviations on bank lending. To avoid such confounders, I end my sample period in 2013. However, my results still hold when using the largest sample I can with the data at hand (until 2018).

⁵I do not have access to the survey, but the consulting firm shared those results in the press (see Gestion, "Tres de cada cuatro empresas fueron afectadas por disparada del dolar." (January 5, 2022), <https://gestion.pe/tu-dinero/tres-de-cada-cuatro-empresas-fueron-afectadas-por-disparada-del-dolar-noticia/>).

(Gilje, Loutskina, and Strahan 2016; Ben-David, Palvia, and Spatt 2017; Slutzky, Villamizar-Villegas, and Williams 2020) focus on reallocation between branches in different geographical locations or shifts from one type of loan to another (Martín, Moral-Benito, and Schmitz 2021). In this paper, I study a different dimension of reallocation, that between business divisions and across different currencies.

This rest of this paper is organized into five sections. Section I reviews the CIP condition and how arbitraging works. Section II describes CIP deviations and the banking system in Peru. Section III describes the data. Section IV presents the methodology and results. Section V concludes.

I. CIP Overview

This section looks at how CIP works and how arbitraging CIP deviations is done. It also introduces definitions I use throughout the paper.

CIP is a nonarbitrage condition. It states that an investor should be indifferent between the following two lending strategies: lend a particular currency directly (i.e., in the spot market, money market, or cash market), or lend it synthetically. Lending soles directly is depicted in red in Figure 1, while lending soles synthetically is shown in blue. *Lending soles synthetically* involves lending dollars and hedging the exchange rate risk. The notes below the figure describe the details of each lending strategy and its returns.

Denote the n -year annualized *soles* (PEN) rate as $y_{t,t+n}$, the n -year annualized *dollar* (USD) rate as $y_{t,t+n}^{\$}$, the *spot rate* at which the bank would buy dollars as S_t and the *forward rate* today at which the bank sells dollars forward at $t + n$ as $F_{t,t+n}$. With this notation, Figure 1 and its description explain how, under CIP, the proceeds at $t + n$ of lending soles directly equal the proceeds of lending soles synthetically:

$$(1) \quad (1 + y_{t,t+n})^n = \underbrace{\frac{F_{t,t+n}}{S_t} \times (1 + y_{t,t+n}^{\$})^n}_{(1 + y_{t,t+n}^{fwd})^n}.$$

The yearly return of lending soles synthetically is the soles synthetic rate (or forward-implied soles rate), denoted as $y_{t,t+n}^{fwd}$. From equation (1), it follows that

$$(2) \quad y_{t,t+n}^{fwd} \equiv \left(\frac{F_{t,t+n}}{S_t} \right)^{1/n} \times (1 + y_{t,t+n}^{\$}) - 1.$$

When there are deviations from CIP, equation (1) does not hold and one lending strategy provides a higher payoff than the other. The difference between the payoffs is known as the *cross-currency basis*, $x_{t,t+n}$. In the literature, the cross-currency basis is typically defined in dollar terms: $x_{t,t+n} = y_{t,t+n}^{\$} - y_{t,t+n}^{\$,fwd}$. Because I draw my analysis in this paper from the Peruvian banks' perspective, I define the cross-currency basis in soles terms:

$$(3) \quad x_{t,t+n} = y_{t,t+n}^{fwd} - y_{t,t+n}.$$

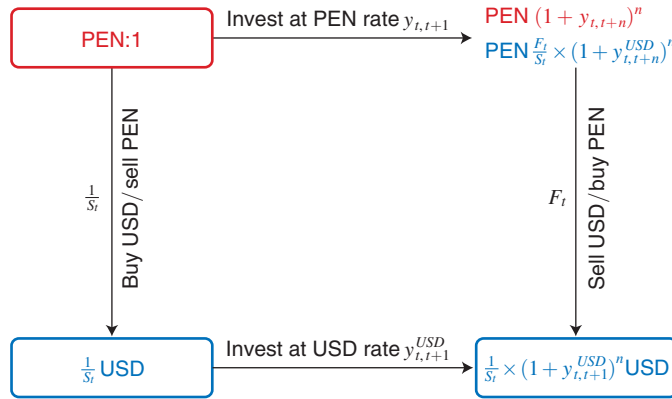


FIGURE 1. EXAMPLE OF COVERED INTEREST RATE PARITY (CIP)

Notes: This figure shows an example of CIP. Here, an investor should be indifferent between two strategies. The first strategy, highlighted in red, is to lend 1 sol (*PEN*) directly in the spot market at the rate $y_{t,t+1}$. When the investor does this, at $t + 1$ the investor will have $PEN 1 + y_{t,t+1}$. The second strategy, highlighted in blue, starts by using the *PEN* 1 that the investor has at time t and changing it for dollars (*USD*). Denoting the exchange rate as S_t *PEN* per *USD*, the investor will have *USD* $\frac{1}{S_t}$. The investor lends these *USD* directly at the *USD* rate of $y_{t,t+1}^{USD}$. Hence, as of $t + 1$, the investor will receive $\frac{1}{S_t} \times (1 + y_{t,t+1}^{USD})$ dollars. CIP means that, as of time t , locking into a $t + 1$ exchange rate to convert the *USD* return into *PEN* should give the same *PEN* as if these *PEN* were lent directly. The $t + 1$ exchange rate at which the investor can lock into in period t is given by the forward exchange rate F_t . Using the F_t exchange rate (also quoted as soles per dollar) to convert the dollar loan proceeds to *PEN*, gives $PEN \frac{F_t}{S_t} \times (1 + y_{t,t+1}^{USD})$. Therefore, under CIP, the soles returns of the red and blue strategies are the same: $1 + y_{t,t+1} = \frac{F_t}{S_t} \times (1 + y_{t,t+1}^{USD})$.

Online Appendix B shows that the definition of the cross-currency basis in dollar terms commonly used in the literature ($x_{t,t+n} = y_{t,t+n}^{\$} - y_{t,t+n}^{\$,fwd}$) is equivalent to equation (3). While I use the non-log form, note that by taking logs to equation (3), the cross-currency basis can also be understood as the difference between the forward premium and the interest rate differential between the soles and dollar money market rates:

$$(4) \quad \log(x_{t,t+n}) = \underbrace{\frac{1}{n}(f_{t,t+n} - s_t)}_{\text{Fwd Premium}} - \underbrace{(y_{t,t+n} - y_{t,t+n}^{\$})}_{\text{Rate Differential}}.$$

During my sample period, the cross-currency basis was mostly negative. This means that when local banks were arbitraging the CIP deviations, banks would be profiting by borrowing synthetic soles and lending them in the money market. When banks borrow soles synthetically, they effectively borrow dollars in the money market, and hedge the exchange rate risk. Hence, arbitraging a negative basis consists of four specific transactions: (i) borrowing dollars in the cash market, (ii) converting these dollars to soles, (iii) lending in soles in the money market while (iv) engaging in a forward contract that sells the soles loan proceeds the bank receives at $t + n$ to convert them to dollars. With the dollars the bank receives from the forward contract, the bank pays back the dollars it borrowed at time t in the money market. What remains as profit, in terms of annualized return, is the cross-currency basis (in absolute terms).

Opposite transactions would be needed to arbitrage the cross-currency basis when it is positive. Such would have been the case out of my sample period, when the cross-currency basis increases. Moreover, because this paper uses the cross-currency basis (in non-log form) as an explanatory variable, the coefficients are interpreted as an increase in the cross-currency basis. Therefore, transactions compatible with banks' arbitraging CIP deviations imply that banks (i) borrow soles in the money market, (ii) convert these soles to dollars in the spot market, (iii) lend dollars in the money market, and (iv) sell the dollars they receive from the amount they lent in the forward market to repay their soles loan.

Intuitively, the sign of the cross-currency can be interpreted as relative scarcity of a currency. The scarce currency is that which the market wants to borrow but not lend. When the cross-currency basis is negative, there is scarcity of dollars (investors want to borrow dollars) relative to soles (investors want to lend soles). Investors selling dollar forward (and driving the forward premia below the rate differential) want a liability in dollars and asset in soles, while arbitrageurs purchasing these forwards need to fund dollars spot to unwind their forward positions by selling dollars spot. The opposite when the cross-currency basis is positive.

II. Setting

This section has three subsections. Section IIA describes the main banking system's business divisions, emphasizing how the trading division works. Section IIB describes the importance of foreign investors in the exchange rate market. Section IIC shows the behavior of CIP deviations in Peru and its similarities with those in other emerging economies.

For brevity, I have placed additional description of the institutional framework and other exchange rate related policies in online Appendix C, which shows that these policies and institutional factors do not affect the results of the paper.

A. Banking System

The financial system in Peru comprises 13 banks and other types of financial institutions, including financial corporations, financial cooperatives known as *cajas*, and leasing companies. These 13 banks hold over 90 percent of the financial system's assets. My results in this paper are independent of whether I include only the 13 banks or the whole financial system. But, because of data limitations and differences in behavior, I focus on banks.⁶

The main business division of banks is household and commercial lending, representing 62 percent of the banking system's assets. Similar to many emerging economies (see Dalgic forthcoming; Montamat 2020; Gutierrez, Ivashina, and Salomao 2023), banks borrow and lend in both dollars and local currency to households and firms. Throughout the sample period, loan dollarization averaged 59 percent, and

⁶ Among the largest four banks operating in Peru, two are domestic and two are branches of foreign banks. For foreign banks to be considered part of the financial system in Peru, they need to have a branch there. This branch would be subject to Peruvian regulation and would have to comply with the same regulatory standards as domestic banks. Regardless of their country of origin, I refer to all banks operating in Peru as "local banks."

deposit dollarization averaged 55 percent. The rest of the economy operates in soles with limited hedging (both naturally and through derivatives, as in Alfaro, Calani, and Varela 2023). In this setting, changes in the share and growth of dollar lending relative to soles have direct consequences for currency mismatches among firms and households.

The other important division is trading, which includes USDPEN exchange rate markets (spot and forward) and investments in fixed income. Banks are market makers in these markets, giving them two natural advantages for arbitraging CIP deviations over banks not operating in Peru. Firstly, they are the only banks able to trade spot transactions in the interbank market. Secondly, they are the only banks with access to the central bank's primary auctions for money market instruments, crucial for trading and arbitraging CIP deviations. When CIP deviations are negative and banks invest in soles assets, key instruments include the central bank's primary auctions for soles certificates of deposit (CDBCRPs) and auctions for soles term deposits at the central bank. Additionally, unlike foreigners (post-2008), local banks face no restrictions when trading CDBCRPs in the secondary market. When CIP deviations are positive and banks require soles funding, the central banks' soles repos, available only to local banks, are very useful.

As is common in emerging economies, regulations limit banks' exposure to exchange rate risk (Canta, Collazos, and Shiva 2007; Tobal 2018; Alfaro, Calani, and Varela 2023). Since exchange rate risk includes both forward and spot transactions, the net of these transactions must be small relative to their equity. However, they can balance a flow in the forward market with one in the spot market and, hence, conduct the transactions required to arbitrage the CIP deviations. Therefore, finding that banks' spot and forward transactions are consistent with arbitraging CIP deviations (as shown in Section IVA) could indicate that banks use CIP deviations to guide their decisions regarding hedging of their exchange rate risk.

Given that the balance sheets of banks do not separate trading from other activities, it is difficult to know which securities are used for funding and investing when trading in the exchange rate market. Understanding which securities banks use is important to do robustness checks with alternative computations of CIP deviations that consider such rates. Hence, I interviewed traders to clarify this. The relevant assets and liabilities coincide with those they report to the bank regulator as short-term assets (less than one year), excluding credit on the asset side, and checking and savings accounts on the liability side. Table A.I of the online Appendix shows the summary statistics for these.⁷

On the liability side, banks use interbank rates as benchmark rates. While interbank loans represent only a small fraction of their overall short-term funding, these rates determine the rates paid for other liabilities (suggested by unreported regressions). Banks complement interbank borrowing with term deposits from institutional

⁷In Peru there are banks that are only trading desks. While these banks are not part of my sample because I look at the interaction between trading and lending, these traders can provide a clear picture of which funding rates and securities to use for arbitraging without the confounder of a different division. I also interviewed traders whose banks have both lending and trading divisions, and while, for the latter, funding is sourced through their Treasury unit, they argue that the rates they receive through transfer pricing from their Treasury unit resemble closely those from securities used in banks with only trading division.

investors and large corporations. On the asset side, the trading division invests in deposits at the central bank and in investment securities.

Soles deposits at the central bank are marginally larger than CDBCRPs, but only a fraction of them could be used for checking accounts. Therefore, in robustness checks that account for the type of investment traders use for arbitrage, I take CDBCRPs as the relevant investment account. Because banks do not have short-term dollar securities, in those robustness checks, I take dollar deposits at the central bank as the main investment dollar account.

B. Foreign Investors and the Exchange Rate Market

This subsection analyzes banks' counterparties in the exchange rate forward market to better understand their trading activities. I focus on the forward market due to a lack of sufficient counterparty information for spot transactions.

I manually classify each counterparty in each forward transaction into broad counterparty-type categories. Foreign investors, who mostly trade nondeliverable forwards, are the main counterparties of banks, accounting for more than 60 percent of the traded volume. In the one-month maturity, trades with foreign investors represent 54 percent of the volume traded. More generally, across all counterparties, the one-month maturity is the most-traded forward contract, representing 43 percent of all forwards traded. The next most heavily traded contract is the two-months forward, representing 13 percent of all forwards traded. Accordingly, I use the one-month forward maturity in my baseline analysis. I use other tenors in my robustness checks, and my results hold.

To arbitrage CIP deviations, banks must have a specific position in the forward market, and foreign investors on the opposite side are not going to arbitrage these deviations (see Table A.III in the online Appendix).

At this point, one might ask, why wouldn't foreign investors arbitrage CIP deviations instead of being on the other side of trades? Two explanations for this are (i) market segmentation and (ii) carry-trade motives.

First, market segmentation between foreign and local banks⁸ sets foreign investors at a disadvantage relative to local banks when trading CIP deviations. Foreign investors do not have access to key money market instruments and the interbank spot exchange rate market that local banks access. Hence, their profitability when arbitraging CIP deviations is often lower than those obtained by local banks.

Second, as I describe in the next subsection, CIP deviations were mostly negative during my sample period. During this time, banks held long dollar forward positions, while foreign investors sold dollars forward. Foreign investors might sell dollars forward to engage in carry trade if they have limited balance sheets to allocate to Peru, rather than purchasing dollars forward to arbitrage CIP deviations. This selling pressure from foreigners in the forward market can decrease the forward premia and generate negative CIP deviations. As I discuss in the next subsection, these negative CIP deviations can be the compensation banks obtain for purchasing dollars forward

⁸ See online Appendix D for more information on the market segmentation.

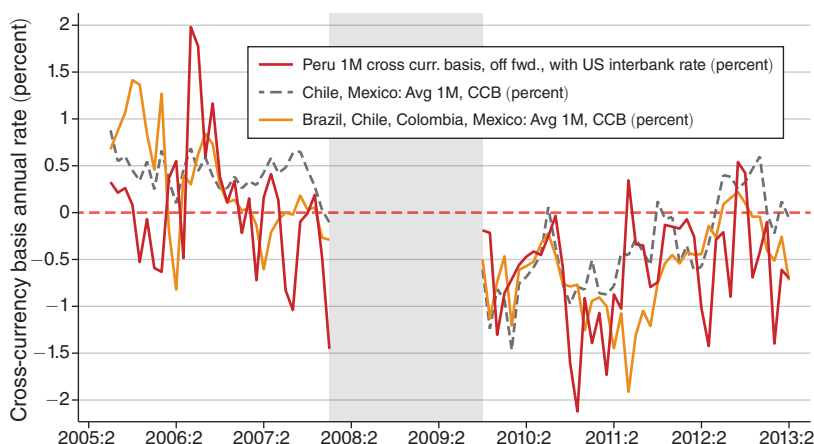


FIGURE 2. CIP DEVIATIONS IN PERU AND OTHER LATIN AMERICAN COUNTRIES

Notes: This figure plots the USDPEN cross-currency basis against the average of the cross-currency basis of other Latin American currency pairs across time. The orange line is the average of the cross-currency basis of Brazil, Chile, Colombia, and Mexico. The dotted gray line is the average of the cross-currency basis of Chile and Mexico. The red line is the cross-currency basis of Peru. Though the level of Peru's cross-currency basis is closer to the average of Brazil, Chile, Colombia, and Mexico, its movements are more correlated to those in Chile and Mexico. All of these cross-currency bases are computed using the local currency against the dollar, and they are all one-month bases. The shaded gray area represents the GFC. I am not showing these months for two reasons: to prevent an outlier period from affecting the results, and because the significant deviations affect the scale, making it hard to understand the scale in normal times.

that foreign investors want to sell, highlighting the close connection between CIP deviations and carry trade motives.

During my sample period, carry trade was particularly profitable. Because it involves getting a liability in a low interest rate currency, which at this time was the dollar, and getting an asset in the high interest rate currency, which at this time was the sol, foreign investors could decide to sell dollars forward to engage in the carry trade.⁹ In line with this, I find that foreign investors' carry trade annualized returns were 6.3 percent while annualized arbitrage profits (computed with CIP deviations using the London inter-bank offered rate (LIBOR) over the same period) were 1.3 percent.¹⁰

C. CIP Deviations

Figure 2 plots the annualized cross-currency basis for one-month contracts for the soles-dollar currency pair (USDPEN) and the average across other Latin American

⁹This strategy could be preferred to a strategy of doing the carry trade by borrowing dollars and investing in soles short-term assets because of the high fees that apply to foreigners when purchasing the central bank CDs (CDBCRPs).

¹⁰To make these calculations comparable, I computed CIP deviations only when they were negative, since this is when carry trade is done. Within my sample period, this was starting in October 2007, excluding the GFC. I also used one-month forwards and one-month CIP deviations (annualized in both cases). To calculate carry trade profits of foreigners, I calculated the profitability of foreign investors' forward trades when they sold forward contracts, as this is likely the most common way foreigners do the carry trade (given the market segmentation).

currency pairs between February 2005 and February 2013, excluding the global financial crisis (GFC). The plots go until February 2013 because after this date, Peru began implementing regulations that affected both bank lending and the forward markets (see online Appendix E). This makes it hard to analyze the effects of arbitraging CIP deviations on bank lending after that date, which is why I end the sample period there.

There are three takeaways from Figure 2. First, these deviations are large in economic terms, even after considering bid-ask spreads. The average of the absolute value of the basis for USDPEN and other Latin American baskets is close to 0.60 percent, far greater than the 0.23 percent bid-ask spreads. Second, the USDPEN cross-currency basis is closely correlated (above 0.4) to the cross-currency basis of other Latin American countries. Finally, while the cross-currency basis switched signs during the sample period, when removing the GFC, the cross-currency basis was mostly negative. In this context, Peruvian banks could profit from borrowing soles synthetically and lending them in the money market.¹¹

Likely, CIP deviations capture pressures coming from the forward market, the spot market, or the money market. However, in Peru, three pieces of evidence suggest that the initial pressure seems to be coming from the forward market rather than the spot market. The spot and money markets then adjust to possible reactions of banks and the central bank. First, the Central Bank Inflation Reports commonly state that the exchange rate movements are driven by flows of foreign investors in the forward market. Second, banks face investors' sales of dollar forwards when the sol appreciates and investors' purchases when the sol depreciates. This direction does not occur in the spot market. Third, the magnitude of the forward flows is typically three times as large as those in the spot market.

Then, analyzing the forward market, I find that banks' net long-forward positions, which capture demand-supply imbalances in the forward market, are strongly negatively correlated with CIP deviations ($\rho = -0.7$). As previously described, foreign investors are key to driving the flows in this market. While during my sample the flows of foreign investors seem aligned with carry trade motives at times when they sell dollars forward, in general, these flows could represent hedging or speculative motives. The degree of each co-varies with global market conditions.

The importance of forward demand aligns with the conclusions drawn by Liao and Zhang (2021), who emphasize that changes in the demand of dollar forwards to hedge currency risk are key to explaining CIP deviations. While CIP deviations could also reflect costs associated with absorbing imbalances in the forward market, I find that changes in the demand of forwards play a more important role. Moreover, as shown in Figure 3, rates adjust to close the cross-currency basis, albeit they do not completely close the gap.

Market segmentation likely helps CIP deviations to capture shifts in demand (see online Appendix D). Since banks have an advantage accessing the markets related to arbitraging CIP deviations, they can charge a markup when absorbing imbalances

¹¹ While the level of CIP deviations is on average negative during my sample period, the changes in cross-currency basis have been both positive and negative. Given that the regression coefficients analyze changes in the cross-currency basis and not the levels, the results discussed when interpreting the regression coefficients of the effects of an increase in the cross-currency basis also pertain to my sample.

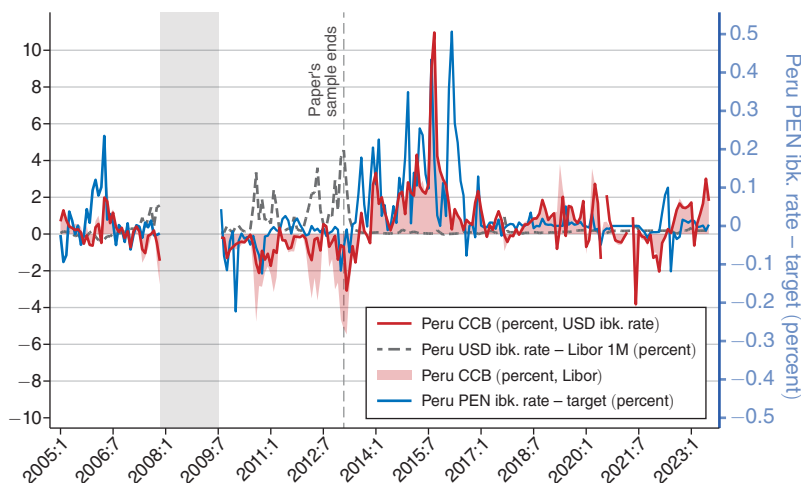


FIGURE 3. CIP DEVIATIONS AND INTEREST RATE SPREADS

Notes: This figure plots Peru's cross-currency computed in two different ways and the interest rate spreads for soles and dollars. The red shaded area shows the cross-currency basis computed using one-month LIBOR. The solid red line is the cross-currency basis computed using dollar interbank rates. I use line as the benchmark for my analysis in this paper. The dollar spread (the gray dotted line) equals the dollar interbank rate in Peru minus the one-month LIBOR. The blue line is the soles interbank spread, computed as the soles interbank rate minus the soles target rate of the central bank. For clarity, I plot the soles interbank spread on the right axis. The sample period is February 2005 to February 2013.

in the forward market. This can explain why CIP deviations existed even before the GFC and before any capital controls that could limit banks' absorption capacity.

III. Data

The sample period for my datasets is February 2005 through February 2013. The end date corresponds with the month in which the Central Bank of Peru began setting regulations that affected both my main outcome and explanatory variables. These regulations, unrelated to Basel III, aimed to reduce dollarization following a significant depreciation after the taper tantrum. They included restrictions on bank lending in dollars (Amado 2022) and on banks' forward transactions (detailed in online Appendix E). The results remain robust when expanding the sample, as shown in last row of Table A2 in the Appendix. Next, I describe the data sources I use.

Market Data.—I obtained market-based data on foreign exchange and money market from Bloomberg (2020). I also obtained local interbank rates from the central banks of Peru, Chile, and Mexico. I used all of these to compute the cross-currency basis across various currency pairs. The summary statistics for the USDPEN cross-currency basis and other currency pairs are reported in Table A1 in the Appendix.

I use the interbank rates to compute the cross-currency basis because (i) traders mention they use these to price the cross-currency basis, (ii) the interbank rates

influence all other rates,¹² and (iii) unlike deposits, interbank loans are typically used only for trading. I do robustness checks with other rates as well.

Additionally, Figure 3 shows that the interbank dollar rate better reflects Peruvian banks' cost of funding than LIBOR. When the cross-currency basis is negative, the dollar interbank rate exceeds LIBOR. Therefore, using the dollar interbank rate yields significantly smaller deviations from cross-currency basis than using LIBOR. However, deviations from CIP remain large even after accounting for the market reaction of rates. Using the dollar interbank rate becomes less important relative to the LIBOR after 2013, when my sample period ends. This is because since 2013, CIP deviations in Peru have mostly been positive; in this case, soles liquidity is scarce (as I discuss in Section IVB).

Bank-Level Data.—I use bank-level data combined from a series of individual bank reports to Superintendencia de Banca y Seguros (SBS), the bank regulator. These reports are mandatory for all banks operating in Peru; most of them are confidential.

To compute the FX side of each bank's arbitrage activities, I combine two reports. The first report entails the universe of their forward contracts. With these contracts, I can compute the net long dollar forward position of a bank, i.e., the net long dollar position of all trades that are currently active (that have not expired).¹³ The second report contains their daily spot transactions, which I use to compute each bank's spot position with a higher frequency than the reported balance sheet position.¹⁴

To calculate the rate side of each bank's arbitrage transactions and to provide robustness checks using alternative measures of CIP deviations, I proceed in two steps. In the first step, I combine two reports to understand the liabilities and investments that are most likely to be used for arbitraging CIP deviations. The first report contains information on money market positions, including banks' interbank loans, financial obligations, investments in short-term assets, and liquidity ratios. This report does not include interest rates. The second report includes banks' assets and liabilities split by maturity.

¹² A 1 percentage point increase in the soles interbank spread (interbank rate minus soles target rate) is associated with a 3 percentage point increase in the one-month term deposit spread, while a 1 percentage point increase in the dollar interbank spread is associated with a 0.65 percentage point increase in the one-month term deposit spread. These numbers correspond to unreported regressions.

¹³ Precisely, the net long-dollar forward position of a bank at time t is

$$Fwd Pos_t = Fwd Pos_{t-1} + Fwd Buy_t - Fwd Sell_t - (Fwd Buy Exp_t - Fwd Sell Exp_t),$$

where $Fwd Pos_t$ is the net long dollar forward position, $Fwd Buy_t$ are the forward dollar purchases, $Fwd Sell_t$ are the forward dollar sales, $Fwd Buy Exp_t$ are the purchases of dollar forwards that expired, and $Fwd Sell Exp_t$ are the sales of dollar forwards that expired.

¹⁴ A bank's long-dollar spot position can be computed in two ways:

$$\begin{aligned} Spot Pos_t &\equiv Spot Pos_{t-1} + USD purchase_t - USD sale_t \\ &\equiv USD Asset_t - USD Liab_t, \end{aligned}$$

where $Spot Pos_t$ is the net long-dollar spot position, $USD purchase_t$ are the dollars purchased in spot, $USD sale_t$ are the dollars sold in spot, $USD Asset_t$ is the dollar assets, and $USD Liab_t$ is the dollar liabilities. The subscript refers to time. I have daily data for each bank's spot purchases and sales, and I have monthly data for each bank's assets and liabilities.

Then, in the second step, I build a dataset on funding and investment rates combining three other reports. For funding rates, I use a first confidential report that includes rates and balances on new deposits, split by deposit type. For investment rates in soles, I use a second dataset that includes monthly snapshots of all fixed-income holdings of banks. This dataset includes information on the security CUSIP, purchase date, currency, yield-to-maturity, and maturity of the securities. Importantly, it has the interest rates and date at which each bank bought the central banks' certificates of deposit. For investment rates in dollars, I use a final dataset that has the daily rates at which banks invest dollars at the central bank; it also has daily interbank rates. I use the combination of funding and investment rates to build robustness checks using bank-level CIP deviations.

Panel A of Table A.II in the online Appendix shows various summary statistics, including those related to "net matched position" and $\hat{\beta}$, which are discussed in Section IVA. An important aspect to highlight from these statistics is that I find significant heterogeneity in the use of these derivatives. Some banks trade no exchange rate forwards or swaps at all, while others trade extensively where the trade volume represents more than 80 percent of their assets.

Bank-Firm-Level Data.—The combination of the previous series of datasets, along with other tests, allows me to present evidence that banks seem to arbitrage CIP deviations, and when doing so, they face scarcity of the currency required to arbitrage. To link these findings with bank lending, I also use the credit register collected by the SBS. It is the most granular dataset on bank loans; together with the spot and forward datasets, it is the main dataset I use in this paper.

This credit register, which is confidential, contains the monthly balances of all commercial loans outstanding in dollars and soles to firms that, during the sample period, had a loan outstanding of more than 300,000 soles (approximately 100,000) in aggregate. I have a total of almost 28,000 firms. Panel B of online Appendix Table A.II shows the summary statistics for these firms.¹⁵

IV. Methodology and Results

This section studies the effect of arbitraging CIP deviations on bank lending. I proceed in three steps. In Section IVA, I show that banks' money market and exchange rate transactions are consistent with arbitraging CIP deviations but that some banks arbitrage more than others. In Section IVB, I show that banks face balance-sheet constraints when arbitraging these deviations, suggesting that arbitraging CIP deviations could be using resources that otherwise would have been used in lending. In Section IVC, I provide evidence showing that arbitraging CIP deviations is associated with changes in lending.

¹⁵ "Small firms" have yearly sales below million soles (approximately \$6.5 million), "medium firms" have yearly sales between 20 million and 200 million soles (\$6.5 million to \$65 million), and "large firms" have yearly sales above 200 soles. These correspond to the medium, large, and corporate categories that the SBS uses to classify firms.

A. Are Banks' Transactions Consistent with Arbitraging CIP Deviations?
Are There Differences across Banks?

To demonstrate that banks' money market and exchange rate transactions are consistent with arbitraging CIP deviations, I test whether the correlations between CIP deviations and the required transactions have the expected signs.

Equations (5a) and (5b), which are aggregate and bank-level estimations respectively, estimate the correlations between CIP deviations and banks' exchange rate and money market transactions. I allow the strength of these correlations to be asymmetric depending on whether the cross-currency basis is positive or negative because the arbitrage requires banks to either borrow or lend a particular currency depending on the sign of the deviations:¹⁶

$$(5a) \quad y_t = \theta_0 + \theta_1 CCB_t \cdot \mathbf{1}\{CCB_t > 0\} + \theta_2 CCB_t \cdot \mathbf{1}\{CCB_t \leq 0\} + \varepsilon_t$$

$$(5b) \quad y_{bt} = \theta_0 + \theta_1 CCB_t \cdot \mathbf{1}\{CCB_t > 0\} + \theta_2 CCB_t \cdot \mathbf{1}\{CCB_t \leq 0\}$$

$$+ \text{Bank FE} + \varepsilon_{bt}.$$

In these equations, CCB_t is the USDPEN cross-currency basis and $\mathbf{1}(\cdot)$ is the indicator function. The dependent variables, y_t and y_{bt} , are money market or exchange rate positions, scaled by total assets. Money market positions include interbank borrowing, obligations with financial institutions and the central bank (two borrowing accounts that are more likely to be influenced by banks' demand of arbitrage funds), investing in the central bank's certificates of deposit (CDs) or sovereign debt, and investing in other bonds. Exchange rate positions include spot and forward. y_t aggregates the data at the month level, while y_{bt} is at the bank-month level. Bank fixed effects (*Bank FE*) are also present in the bank-level regression. The coefficients of interest, θ_1 and θ_2 , capture the correlations between y_t and the cross-currency basis when it is positive and when it is negative, respectively.

Table 1, panel A shows that the aggregate-level correlations between CIP deviations and banks' exchange rate and money market transactions (equation (5a)) are statistically significant and have the expected signs. As the cross-currency basis increases, borrowing soles and lending them synthetically becomes more profitable. In line with the arbitrage profitability, we see that banks (i) increase their borrowing in soles and (ii) increase their lending of soles synthetically.¹⁷ Asymmetry is also present and in the expected direction, in terms of both magnitude and statistical significance. When the cross-currency basis is positive, banks borrow more in soles than when it is negative. Banks also lend soles synthetically: (i) buy more dollars spot, (ii) sell more dollars forward, and (iii) invest more in dollars when the cross-currency basis is positive than when it is negative.

¹⁶For example, because the cross-currency basis is positive when the implied soles rate is higher than the soles cash rate, an arbitrageur is likely to increase soles cash borrowing when the cross-currency basis is positive, compared to when the cross-currency basis is negative.

¹⁷As per Section I, lending soles synthetically means that banks lend in dollars but hedge the exchange rate risk (i.e., they buy dollars spot to lend dollars in cash but sell dollars forward).

TABLE 1—EVIDENCE CONSISTENT WITH ARBITRAGE OF CIP DEVIATIONS

	<i>Panel A. Aggregate</i>		<i>Panel B. Bank-level</i>		<i>Panel C. High-arbitrage</i>		<i>Panel D. Low-arbitrage</i>	
	<i>CCB>0</i>	<i>CCB<0</i>	<i>CCB>0</i>	<i>CCB<0</i>	<i>CCB>0</i>	<i>CCB<0</i>	<i>CCB>0</i>	<i>CCB<0</i>
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Borrowing</i>								
Interbank loans								
1. <i>PEN</i>	0.37 (0.15)	0.05 (0.05)	0.35 (0.12)	0.08 (0.05)	0.66 (0.17)	0.16 (0.08)	−0.04 (0.09)	−0.01 (0.05)
2. <i>USD</i>	−0.04 (0.03)	−0.07 (0.04)	−0.02 (0.05)	−0.08 (0.08)	0.00 (0.09)	−0.15 (0.14)	−0.05 (0.02)	−0.00 (0.02)
Financial obligations								
3. <i>PEN</i>	1.22 (0.47)	−0.29 (0.17)	0.46 (0.40)	−0.29 (0.14)	1.20 (0.43)	−0.23 (0.13)	−0.46 (0.41)	−0.38 (0.24)
4. <i>USD</i>	−2.44 (0.82)	−2.98 (0.66)	−1.11 (0.34)	−0.87 (0.27)	−2.21 (0.66)	−2.05 (0.47)	0.27 (0.42)	0.55 (0.27)
<i>Currency exchange</i>								
Spot position								
5.	4.21 (0.60)	2.61 (0.48)	3.48 (0.58)	2.20 (0.46)	5.42 (0.93)	3.74 (0.77)	1.05 (0.20)	0.32 (0.15)
Forward and swap position								
6.	−3.55 (0.54)	−2.06 (0.41)	−2.48 (0.51)	−1.87 (0.42)	−4.31 (0.89)	−3.36 (0.73)	−0.18 (0.06)	−0.06 (0.09)
<i>Lending</i>								
Central bank and government								
7. <i>PEN</i>	−2.61 (1.23)	0.37 (1.11)	−1.76 (0.73)	−0.08 (0.94)	−2.52 (1.11)	−0.22 (1.17)	−0.77 (0.36)	0.12 (0.68)
8. <i>USD</i>	0.18 (0.15)	0.30 (0.12)	0.34 (0.15)	0.27 (0.08)	0.04 (0.22)	0.31 (0.14)	0.74 (0.09)	0.23 (0.07)
Investments								
9. <i>PEN</i>	−1.09 (1.09)	0.22 (1.21)	−0.83 (0.70)	−0.36 (1.08)	−1.22 (1.00)	−0.64 (1.42)	−0.35 (0.40)	−0.03 (0.69)
10. <i>USD</i>	1.34 (0.31)	0.67 (0.21)	1.02 (0.35)	0.48 (0.15)	0.80 (0.33)	0.54 (0.17)	1.28 (0.41)	0.41 (0.16)
Observations*	77	77	873	873	479	479	394	394

Notes: This table shows the results of estimating linear regressions of the different accounts used for CIP arbitrage on Peru's cross-currency basis, separated by the months when it was positive and negative. For each panel, each row is a regression, where the dependent variable is shown on the left. Variables are written as a percentage of total assets (0–100 scale). The explanatory variable is given in the column subhead. Each regression has two explanatory variables: (i) the cross-currency basis when positive and (ii) when negative, that correspond to the first two columns within each panel. Regressions in panel A were estimated using the monthly time series of aggregate accounts of banks' balance sheets. Regressions in panel B were estimated using data at the bank-month level, with bank fixed effects. Regressions in panel C were estimated restricting the sample of panel B to the banks that arbitrage the most. Panel D covers banks that arbitrage the least. All regressions have been estimated over a sample between February 2005 and 2013 (excluding the global financial crisis (GFC)). In all panels, HAC standard errors were used, allowing for 3-month autocorrelation. In addition, standard errors in panels B, C, and D were clustered by month. Standard errors are in parentheses.

*Central bank and government variables, rows 7 and 8, have small differences in observations for bank-level regressions.

The magnitude of the spot and forward coefficients (Table 1, rows 5 and 6) are worth highlighting. In absolute terms, they are two to three times as large as those in the borrowing and lending sides. This means that banks are funding only part of their dollar purchases with new soles borrowed with interbank loans and financial obligations, as is the case with dollar borrowing when the cross-currency

basis decreases. Accordingly, banks will need to fund their dollar purchases as the cross-currency basis increases through other sources, which can include reducing funding in different business divisions (i.e., previous investments as well as commercial and personal lending).

Shifting to the bank-level results, panel B of Table 1 shows that the transactions are still consistent with arbitraging CIP deviations, but the results are less robust than the aggregate estimations. This is expected if banks' arbitraging activities are heterogeneous and not all banks arbitrage CIP deviations.

To further analyze the differences in banks' sensitivity to arbitrage CIP deviations, I compute bank-level sensitivities of the share of the banks' assets likely used to fund arbitrage after a change in the cross-currency basis. To do so, I first construct a bank-level measure that proxies for the share of a bank's assets invested in arbitraging the cross-currency basis. Then, I use this measure to compute bank-specific sensitivities.

Construction of the Arbitrage Proxy.—To construct the proxy for the share of a bank's assets invested in arbitraging the cross-currency basis, I compute a daily measure of a bank's forward and swap positions that are offset by its spot position.¹⁸ The amount of a bank's short forward position that is effectively matched with its long spot positions is the proxy. I call this variable the *matched position* of a bank. This measure can capture an arbitrage position because any CIP arbitrage position requires banks to offset their forward positions with spot positions. Although banks borrow and lend as part of arbitraging CIP deviations, I rely only on the exchange rate positions, because they are a cleaner proxy than the bank's use of the money market since there is a higher degree of uncertainty on which accounts are used for the borrowing and investing legs of arbitrage. Formally, I define the matched position of a bank as follows:

$$(6) \quad Matched_{bt} = \begin{cases} -\min\{|Spot\ Pos. |, |Fwd+Swap\ Pos. |\}, & \text{if } Fwd+Swap\ Pos. > 0 \wedge Spot\ Pos. < 0 \\ +\min\{|Spot\ Pos. |, |Fwd+Swap\ Pos. |\}, & \text{if } Fwd+Swap\ Pos. < 0 \wedge Spot\ Pos. > 0 \\ 0, & \text{if } \text{sgn}(Fwd+Swap\ Pos.) = \text{sgn}(Spot\ Pos.). \end{cases}$$

Because the matched position of a bank ($Matched_{bt}$ in equation (6)) is the amount of a bank's short dollar forward position that is offset with long dollar spot positions, it is computed as the minimum between the absolute value of the spot position and the forward position. $Matched_{bt}$ is negative when a bank has a net long dollar forward position that is offset and a net short dollar spot position (the first case in equation (6)), and positive when the converse occurs (the second case). When banks do not offset spot positions with forward positions, they are not arbitraging, so $Matched_{bt}$ is zero.

¹⁸I compute positions (that is, stock holdings rather than flows) for two reasons. First, I want to associate changes in the cross-currency basis with changes in these positions. This is given by the coefficient in the regression between these positions and the cross-currency basis. Second, because when a forward contract expires, a bank needs to renew it to keep its spot position hedged with the forward position. The purchases and sales of forward contracts could not be representative of whether a bank is arbitraging, as these could be due to renewals of expiration of forward contracts. In any case, because banks need to hedge their forward positions with their spot positions, when a bank decides not to renew a forward position, it will also need to change its spot position.

Computation of Bank-Specific Sensitivities.—I use $Matched_{bt}$ to estimate β , the measure I use to compare banks' sensitivities to arbitrage. I estimate β separately for each bank by using the following time-series regression:

$$(7) \quad \left(\frac{Matched}{Assets} \right)_{bt} = \alpha_b + \beta_b CCB_t + \varepsilon_{bt} \quad \forall b \in B,$$

where t indexes months, b indexes a particular bank, and B is the set of all banks in the sample. Month-level variables were calculated as the averages of their daily counterparts.

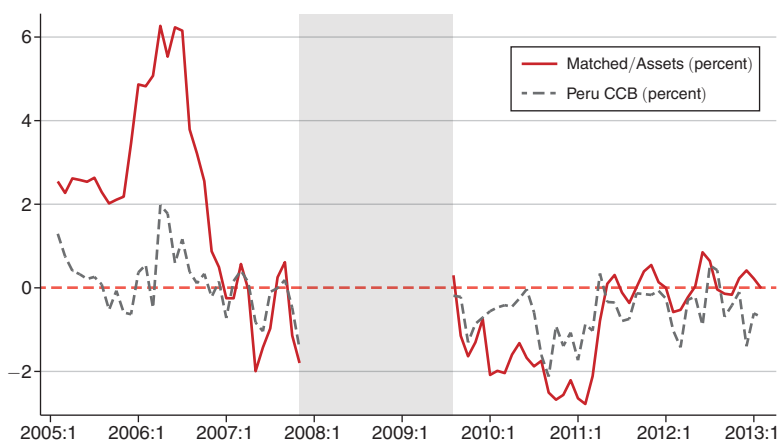
Because $\hat{\beta}$ measures the correlation between $Matched_{bt}$ and the cross-currency basis, we expect this coefficient to be positive when banks arbitrage CIP deviations. This is because when CIP deviations are positive, the arbitrage involves buying dollars spot and selling dollars forward (i.e., positive “ $Matched_{bt}$ ”) given that the soles synthetic rate would be higher than the cash rate. Indeed, as shown in Figure 4 (which plots the aggregate matched position of the banking system scaled by total assets against the cross-currency basis), the aggregate matched position of the banking system is highly positively correlated ($\rho = 0.70$) with the cross-currency basis.

At the bank level, we should observe that $\hat{\beta}_1 > \hat{\beta}_2 > 0$ if Bank 1 pursues a more aggressive arbitrage strategy than Bank 2. In this case, Bank 1 matches a higher percentage of its assets in the direction predicted by arbitrage when the cross-currency basis changes by 1 percentage point. Consequently, I interpret the estimated $\hat{\beta}_b$ coefficient as proxy of bank b 's intensity of arbitrage sensitivities/activities.

Estimating equation (7) separately for each bank yields considerable heterogeneity in the resulting coefficients. A concentration of banks shows near-zero $\hat{\beta}$ s (low-arbitrage banks), whereas another group of banks has $\hat{\beta}$ s that are much larger than or significantly different from zero (high-arbitrage banks). The estimated coefficients of the low- and high-arbitrage banks lie in approximate ranges of $[0, 0.2]$ and $[1.6, 4.8]$, respectively.

I find that $\hat{\beta}$ s proxy banks' arbitrage activity well. The exchange rate and money market transactions of banks that arbitrage more (higher $\hat{\beta}$) are more consistent with arbitraging CIP deviations than those that arbitrage less. Panels C and D in Table 1 show the results of splitting high- from low-arbitrage banks and estimating the same regressions as those in panel B for each group. As expected, the estimated coefficients for the arbitrage accounts are larger in the group of high-arbitrage banks than they are in the low-arbitrage group. Specifically, the coefficients for high-arbitrage banks (panel C) are, generally, very consistent with banks that are using these accounts for arbitrage, in terms of sign, significance, and asymmetry. However, the coefficients for the low-arbitrage banks (panel D), are either (i) opposite to arbitrage, or (ii) nonsignificant, or (iii) smaller than their counterparts from panel C.

Explanations for Arbitrage Heterogeneity.—An underlying question is why some banks are able to arbitrage more than others? Online Appendix F addresses this question by providing correlations to show suggestive evidence of potential

FIGURE 4. CIP DEVIATIONS AND *MATCHED/ASSETS*

Notes: This figure plots Peru's cross-currency basis juxtaposed with a proxy of the share of assets allocated to arbitraging such deviations, "*Matched/Assets*," for the total Peruvian banking system. The sample period is February 2005 to February 2013.

explanations. This analysis, however, has an important limitation: explaining the cross-section of banks is difficult because there are few banks.

With this caveat in mind, online Appendix F evaluates two groups of possibilities. The first is that some banks can arbitrage more than others because they have fewer balance-sheet constraints. The second is that banks that arbitrage more do so because they can more easily unwind positions in the forward market. This can happen when banks that arbitrage more are those that have a greater set of clients, with flows that offset the trends in the forward market.

In sum, it seems that differences in banks' arbitrage sensitivities could be explained by their client heterogeneity. Banks arbitrage more when they can count on their clients to help them unwind positions. I also find that banks that arbitrage more have higher liquid assets in line with balance-sheet constraints explaining part of the heterogeneity. However, because the analysis is subject to important reverse-causality concerns, making conclusions about the relationship between such constraints and arbitrage is difficult.

B. *Is the Currency Needed to Arbitrage CIP Deviations Scarce?*

This section examines whether the currency that banks need to borrow to arbitrage is scarce when CIP deviations exist. If this is the case, banks are allocating a scarce resource to arbitraging CIP deviations. Therefore, the arbitrage can affect funding of that currency in other business divisions, such as commercial lending.

For example, when the cross-currency basis is positive (soles synthetic rate is higher than the soles cash rate), banks arbitraging these deviations need to borrow soles cash. Banks can source funds internally or externally. On one hand, if banks choose to source funds internally, they will be reallocating soles away from other divisions, such as the lending division and soles lending falls. On the other hand, if banks source externally, they need to pay more to borrow soles, likely resulting in

higher soles lending rates and inducing firms to substitute their soles borrowing for dollar borrowing. The converse happens when the cross-currency basis is negative.¹⁹

The behavior of the soles and dollar interbank spreads (with respect to the target rate) during positive versus negative cross-currency basis indicates suggests scarcity in the currency required to arbitrage. Figure 3 shows that the soles interbank rate increases above the central bank's target rate when the cross-currency basis is positive and arbitraging CIP deviations requires banks to borrow soles. Similarly, the dollar interbank rate increases above the LIBOR when the cross-currency basis is negative and arbitraging CIP deviations requires banks to borrow dollars.²⁰ Figure 3 plots these correlations through 2023; it shows that these correlations hold generally.

More formally, to present evidence that banks seem to face borrowing constraints in the currency required to arbitrage, I replicate the regressions of the previous section (equations (5a) and (5b)) using interest rate spreads and liquidity ratios. I use interest rate spreads (the rate minus monetary policy rate) so as not to pick up changes in monetary policy. Table 2 presents the results in three levels: aggregate, bank-level, and decomposed by high- and low-arbitrage banks.

The aggregate and bank-level results are similar: they show a scarcity of the currency required to arbitrage. A 1 percentage point increase in the cross-currency basis is associated with a 0.3 percentage point increase in the soles term deposit spread and a 2 to 3 percentage point decrease in the share of soles liquid assets. Analogously, a 1 percentage point decrease in the cross-currency basis is associated with an increase of 0.43 to 0.56 percentage points in the dollar spread and a decrease of 1.51 to 4.41 percentage points in the share of dollar liquid assets.

The demand for funds to arbitrage CIP deviations could explain the scarcity of the currency required to arbitrage, but because CIP deviations correlate with other macroeconomic factors, the relationships above are not causal. The origin of the scarcity of liquidity, however, is not relevant for this paper. What matters is that the currency required to arbitrage CIP deviations is scarce when banks want to arbitrage. This means that banks are optimizing under funding constraints: to arbitrage CIP deviations, they need to reallocate funds internally or pay more to obtain funds externally—either way can impact bank lending. I will address this in the next section.

C. How Does Arbitraging CIP Deviations Affect Bank Lending in Soles and Dollars?

This section examines whether the arbitrage of CIP deviations in a context where the funding currency is scarce can affect bank lending in soles and in dollars.

Estimating the effect of arbitraging CIP deviations on bank lending is challenging, for three main reasons. First, CIP deviations are affected by macroeconomic

¹⁹Unfortunately, I lack information about interest rates and tenor at the loan level to test whether the adjustment is through quantities or rates. Therefore, for the bank-lending results in Section IVC, I can use only quantities. However, quantities may just be reflecting worse rates in one currency versus the other. Because client portfolios (and hence client risk) can differ from bank to bank and loans could have various tenors, it is hard to assess the rate adjustment without loan-level rates.

²⁰The positive correlation between the cross-currency basis and the soles spread, as well as the negative correlation between the basis and the dollar spread, occurs despite the CIP deviations having already been computed with interbank soles and dollar rates. Therefore, although these changes in interbank rates reduce the cross-currency basis gap, the CIP deviations are much larger than the changes in rates.

TABLE 2—EVIDENCE CONSISTENT WITH LIQUIDITY PROBLEMS RELATED TO CIP DEVIATIONS

	Panel A. Aggregate		Panel B. Bank-level		Panel C. High-arbitrage		Panel D. Low-arbitrage	
	<i>CCB>0</i>	<i>CCB<0</i>	<i>CCB>0</i>	<i>CCB<0</i>	<i>CCB>0</i>	<i>CCB<0</i>	<i>CCB>0</i>	<i>CCB<0</i>
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Term deposit spread</i>								
1. <i>PEN</i>	0.36 (0.09)	0.25 (0.08)	0.40 (0.12)	0.26 (0.11)	0.41 (0.09)	0.22 (0.08)	0.40 (0.17)	0.31 (0.15)
2. <i>USD</i>	−0.43 (0.15)	−0.56 (0.14)	−0.82 (0.19)	−0.49 (0.13)	−0.68 (0.15)	−0.50 (0.13)	−0.98 (0.25)	−0.49 (0.16)
<i>Interbank loan spread</i>								
3. <i>PEN</i>	0.04 (0.03)	0.02 (0.01)	0.06 (0.04)	−0.00 (0.02)	0.06 (0.05)	0.00 (0.02)	0.07 (0.04)	−0.00 (0.02)
4. <i>USD</i>	−0.42 (0.19)	−0.64 (0.19)	−0.53 (0.23)	−0.60 (0.22)	−0.50 (0.22)	−0.62 (0.20)	−0.60 (0.27)	−0.56 (0.26)
<i>Liquidity ratios</i>								
5. <i>PEN</i>	−3.13 (1.09)	−2.05 (0.55)	−2.59 (0.63)	−1.97 (0.53)	−3.64 (0.98)	−2.96 (0.78)	−1.27 (0.52)	−0.78 (0.44)
6. <i>USD</i>	4.41 (0.65)	1.51 (0.46)	2.66 (0.44)	0.55 (0.40)	3.57 (0.65)	0.81 (0.64)	1.53 (0.35)	0.24 (0.18)
Observations*	77	77	873	873	479	479	394	394

Notes: This table shows the results of estimating linear regressions of different proxies for liquidity constraints on Peru's cross-currency basis, separated by the months when it was positive and negative. For each panel, each row is a regression, where the dependent variable is shown on the left. Variables are written as a percentage of assets (0–100 scale) or as percentage points, if they are interest rate spreads. The explanatory variable is given in the column subhead. Regressions in panel A were estimated using the monthly time series of aggregate accounts of banks' balance sheets. Regressions in panel B were estimated using data at the bank-month level and with bank fixed effects. Regressions in panel C were estimated restricting the sample of panel B to the banks that arbitrage the most. Panel D covers the subsample corresponding to the banks that arbitrage the least. All regressions have been estimated over a sample between February 2005 and 2013 (excluding the GFC). In all panels, HAC standard errors were used, allowing for 3-month autocorrelation. In addition, standard errors in panels B, C, and D were clustered by month. Standard errors are in parentheses.

*There are small differences in observations for bank-level regressions.

shocks that affect CIP deviations and banks' decisions to lend in different currencies. Second, these macroeconomic shocks also affect firms' investment opportunities and thus their credit demand. Therefore, controlling for the effect of these shocks is crucial, both from the bank side and from the firm side. Third, banks' lending decisions themselves can affect CIP deviations. Given that banks operate in the exchange rate and commercial lending markets, their actions affect both markets. A bank that decides to lend in a particular currency and simultaneously hedge the exchange rate risk could change its demand in the forward market and ultimately affect the USDPEN cross-currency basis.

The main regression specification, shown in equations (8a) and (8b)), addresses these problems in two ways. First, it compares how banks with different sensitivities to arbitrage CIP deviations ($\hat{\beta}$) change their lending in dollars and soles on the same month and to the same firm (70 percent of firms have multiple bank relationships) following changes in the cross-currency basis. Comparing across banks on the same month controls for macroeconomic shocks when shocks affect all banks equally. Additionally, performing a within-firm-month analysis (i.e., using firm-month fixed effects) and only comparing changes of bank lending to the same firm reduces concerns that the results could be driven by changes in firms' credit demand.

Second, it instruments the CIP deviations in Peru with those in Mexico and Chile. Using the cross-currency basis of Mexico and Chile as an instrumental variable (IV) not only reduces the influence of shocks to the Peruvian economy on the estimation results but also prevents the results from being biased by Peruvian banks' trading decisions in the exchange rate market that affect the USDPEN cross-currency basis. More precisely, I estimate the following baseline two-stage least squares model:

$$(8a) \quad CCB_{t-1}^{Peru} \times \hat{\beta}_b = \gamma_0 + \gamma_1 CCB_{t-1}^{ChMex} \times \hat{\beta}_b + X'_{b,t-1} \Theta + \psi_b + v_{b,t-1}$$

$$(8b) \quad y_{bft} = \alpha_0 + \alpha_1 \overline{CCB_{t-1}^{Peru} \times \hat{\beta}_b} + \psi_{bf} + \psi_{ft} + X'_{b,t-1} \Psi + \epsilon_{bft}$$

where y is the observed credit outcome (log of USD, PEN, total, and share of USD loans)²¹ given by bank b to firm f on month t ; CCB_{t-1}^{Peru} is the one-month lagged cross-currency basis of USDPEN; CCB_{t-1}^{ChMex} is the average one-month lagged cross-currency basis of Chilean and Mexican peso against the dollar (USDCLP and USDMEX, respectively); $\hat{\beta}$ (estimated in Section IVA) measures the bank arbitrage intensity level; $X_{b,t-1}$ is a vector of one-month lagged bank controls; and ψ_b , ψ_{bf} , and ψ_{ft} refer to bank fixed effects, bank-firm fixed effects, and firm-month fixed effects, respectively. Equations (8a) and (8b) refer to the first and second stages of the model, respectively.

I use the average between the one-month USDCLP and USDMEX cross-currency bases for two reasons. First, for the instrument to be valid, it must be highly correlated with the USDPEN cross-currency basis. The USDCLP and USDMEX cross-currency bases are the two Latin American currencies whose correlation with the USDPEN cross-currency basis is the strongest. The average combined basis of USDCLP and USDMXN has a correlation of 0.54 with the USDPEN cross-currency basis. Second, fewer than 1.1 percent of all of the forward contracts that banks in Peru traded were USDMXN or USDCLP and therefore they are unlikely to affect their prices.

I add bank-firm fixed effects to control for time-invariant characteristics between a bank and a firm. They also control for time-invariant differences across banks. This is important because shocks that correlate with CIP deviations may not affect all banks in the same way. If these shocks are also correlated with banks' sensitivities to arbitrage, the results on bank lending may be driven by the shock that correlates with CIP deviations rather than arbitraging CIP deviations by banks.

Controlling for time-invariant characteristics of banks as well as their relationships with firms helps mitigate this concern. Because the fixed effects do not capture the time-varying component of banks' characteristics, I also add lagged time-varying bank controls. These controls include soles and dollar deposits scaled by total assets,

²¹ For this regression, I sum across all types of loans for each bank-firm-month. Each observation includes only firms that have positive total credit with a bank. However, a firm could be borrowing only soles or only dollars at one point in time. To keep the same number of observations between soles and dollar loans and to avoid considering different samples of firms when looking at soles versus dollar loans, before taking logs I add 1 sol (approximately 0.33 dollars) to all loan balances. Moreover, to make loan balances compatible across time, the dollar loan balances use a constant exchange rate as of the start of the sample, February 2005. I have also performed robustness checks where I do not add 1 sol to the loans. The conclusion remains, although the coefficients are smaller. Similarly, I have also done robustness checks without adjusting the exchange rate to have a constant exchange rate. The results remain.

TABLE 3—FIRST-STAGE RESULTS

	(1)	(2)	(3)
$CCB_{t-1}^{Chile,Mex} \times (\hat{\beta})$	0.811 (0.149)	0.591 (0.136)	0.576 (0.137)
<i>Bank controls</i>	No	No	Yes
<i>Bank fixed effects</i>	No	Yes	Yes
<i>F</i>	29.45	18.77	17.77
Observations	1,348,040	1,348,040	1,348,040

Notes: This table presents the first-stage results for three alternative specifications. They all show the relationship between the USDPEN cross-currency basis and the average basis of USCLP and USDMXN and have been estimated using alternative specifications of equation (8a). The dependent variable for all specifications is $CCB_{t-1}^{Peru} \times (\hat{\beta})$. Both the USDPEN and the average of USDCLP and USDMXN are expressed on a 0–100 scale. Column 1 has no bank controls and no bank fixed-effects. Column 2 adds bank controls only. Column 3 includes bank controls and bank fixed effects. Column 3 corresponds to the first stage of the baseline specification (equation (8a)). The *F*-statistic is that of the first stage, given by Kleibergen-Paap rk Wald *F*-statistic. Standard errors are in parentheses. Standard errors are those from the joint estimation with the second stage, which are clustered by date and firm. The sample period goes from February 2005 to February 2013, excluding the GFC.

log of total assets, return over assets, and share of liquid assets in soles and dollars. Regressions without lagged bank controls yield very similar results.

The coefficient of interest, α_1 , measures the percentage increase in bank lending of increasing arbitrage intensity by 1 (i.e., increasing $\hat{\beta}$ by 1) after a 1 percentage point increase in the cross-currency basis when lending to the same firm in the same month.

Table 3 shows the first-stage results for various specifications, including the baseline specification (column 3). The instrument is statistically significant and stable across specifications. Its strong correlation with USDPEN cross-currency basis also indicates the absence of a weak-instrument problem.

Table 4 shows the second-stage results for the IV baseline specification using five different dependent variables: *log of dollar loans*, *log of soles loans*, *log of total loans*, *the share of dollar loans*, and *the difference between log of dollar loans and log of soles loans*.

The main takeaway from estimating the baseline regression is that an increase in the cross-currency basis increases the share of dollar lending relative to soles lending. That is, banks change the distribution of dollars versus soles lending without affecting total lending much. These results are all significant at 1 percent level. They are also consistent across alternative specifications. Banks that allocate 1 percentage point more of their assets to arbitraging a 1 percentage point increase in the cross-currency basis increase their dollar lending by 11 to 41 percentage points relative to their soles lending. In terms of standard deviations, a 1 standard deviation increase in the cross-currency basis makes banks that respond by arbitraging one-standard-deviation more increase dollar lending relative to soles lending by 71 percentage points, making the share of dollar borrowing increase by 2.48 basis points.

Decomposing this result into soles borrowing and dollar borrowing, I find this result is driven not only by an increase in dollar lending but also by a decrease in soles lending. The range depends on the sample. I obtain the most conservative result, 11 percent, when including only firms that were already borrowing in dollars

TABLE 4—EFFECT OF ARBITRAGING CIP DEVIATIONS ON BANK LENDING: BASELINE SPECIFICATION

	IV				
	$\log(PEN)$	$\log(USD)$	$\log(Total)$	Ratio	$\log(USD) - \log(PEN)$
$CCB_{t-1}^{Peru} \times (\hat{\beta})$	-25.18 (7.312)	16.16 (4.620)	3.085 (1.525)	1.452 (0.426)	41.35 (11.10)
<i>Firm × Month fixed effects</i>	Yes	Yes	Yes	Yes	Yes
<i>Bank × Firm fixed effects</i>	Yes	Yes	Yes	Yes	Yes
Bank controls	Yes	Yes	Yes	Yes	Yes
Firm cluster	18,374	18,374	18,374	18,374	18,374
Month cluster	77	77	77	77	77
Observations	1,348,040	1,348,040	1,348,040	1,348,040	1,348,040

Notes: This table presents the baseline results of the effect of arbitraging CIP deviations on bank lending. The specification is given by equation (8b). The five columns show the IV estimates. The dependent variables in logarithm have been multiplied by 100. The ratio of dollar to total loans is expressed on a 0–100 scale. Standard errors are in parentheses. Standard errors are those from the joint estimation with the first stage. These have been clustered by date and firm. The sample period goes from February 2005 to February 2013, excluding the GFC. To prevent the results of the dollar loans from reflecting changes in the exchange rate, the dollar loans have been converted to soles using a constant exchange rate (corresponding to February 2005).

and soles (row 5 of Table A2 in the Appendix.). I get the largest result when including all firms. My benchmark includes all firms in all regressions. In net terms, my results represent a change in the currency denomination of the loans and a small change in total loans. Therefore, it highlights that the mechanism in this paper involves recomposing banks' balance sheets rather than expanding them.

The analogous OLS regression for all the columns reported in Table 4 is reported in Table A.IV in the online Appendix. Both types of model, OLS and IV, show the same pattern and statistical significance, but the differences between OLS and IV show a consistent negative bias. The bias is as expected and can be explained as follows. A bank that decides to lend more dollars, to meet regulations, will need to hedge. Unless the bank borrows and lends in the same currency, the bank will need to hedge by selling dollars forward. As a market maker, when the bank sells dollars forward, it will set downward pressure to the forward outright ($F_{t,t+n}$ in equation (1)) and decrease the cross-currency basis. This ultimately leads to a lower cross-currency basis, higher dollar lending, and lower soles lending (if lending more in dollars means banks prefer to lend less in soles), and hence, goes against finding a result through the mechanism proposed in this paper. Thus, as expected, OLS estimates are significantly lower than the IV estimates.

D. Robustness

The baseline regression specification shown in the previous subsection addresses various concerns, but some concerns might linger. This section provides robustness checks for the following concerns: (i) the correlation between the basis and other macroeconomic shocks, (ii) the heterogeneity across banks, (iii) the effect of calculating CIP deviations using different variables and tenors, and (iv) the effect of calculating CIP deviations at the bank level instead of at the market level. Online Appendix G adds descriptions on additional robustness checks to concerns regarding (i) the demand

effect coming from the relationship between the exchange rate and foreign trade, (ii) the role played by different types of loans, (iii) the effect of alternative arbitrage intensities, and (iv) the effects of different clustering of standard errors. My results are robust to all these tests. While (i) and (ii) are described in more detail in online Appendix G, (iii) and (iv) are detailed in online Appendix Tables A.VII and A.VIII, respectively.

The Exchange Rate.—One concern is that macroeconomic shocks that correlate with CIP deviations (including changes in the exchange rate) could themselves explain my results. In this section, I focus on the main known macroeconomic correlation: the one between CIP deviations and exchange rate (Avdjiev et al. 2019). In online Appendix C, I show that policy changes do not affect my results.

Figure 5 shows that soles (PEN) depreciate as the cross-currency basis increases. This positive correlation between the exchange rate and the cross-currency basis can confound the effects of arbitraging CIP deviations. Through independent channels, a depreciation of the local currency and an increase in the cross-currency basis can both generate an excess supply of dollar funding and a shortage of local currency funding provided to banks. As the sol depreciates, households and firms will prefer to switch their savings from soles to dollars. Through this channel, which I refer to as the exchange rate channel, as the sol depreciates, banks would increase dollar lending and decrease soles lending to mirror what is happening to their funding side. Although the net effect on bank lending is uncertain (because households and firms will probably demand more soles borrowing as the sol depreciates) if the bank supply side dominates, it is possible the baseline results are picking up the correlation with the exchange rate rather than arbitraging CIP deviations.

However, for the exchange rate channel to threaten the results, it is not enough for it to be correlated with the cross-currency basis. To invalidate the results, because the estimation relies on comparing bank lending across banks with varying sensitivity to arbitrage, we also need the exchange rate channel to have a greater effect on those banks with a higher sensitivity to arbitrage.

To check whether banks that arbitrage more are the more affected by the exchange rate channel, I compare the bank-level sensitivity of bank deposits after changes in the exchange rate with the bank-level arbitrage intensity. I use the sensitivity to bank deposits because this would be the direct channel through which the exchange rate affects banks' liquidity. To compute this sensitivity, I estimate the following time-series regression separately for each bank:

$$(9) \quad \left(\frac{\text{Deposits}}{\text{Assets}} \right)_{bt} = \alpha_b^0 + \alpha_b^1 \log(FX)_t + \varepsilon_{bt} \quad \forall b \in B,$$

where the numerator of the dependent variable is either deposits in soles, deposits in dollars, or total deposits; $\log(FX)$ is the log of the exchange rate.

I find that the deposit sensitivity to the exchange rate does not have the strongest impact on the banks that arbitrage the most. Table A3 in the online Appendix shows the summary statistics of the estimated coefficients, splitting banks into three groups, depending on their arbitrage intensity. The medium arbitrage-intensive banks are those for which dollar deposits increase the most when the sol depreciates. For

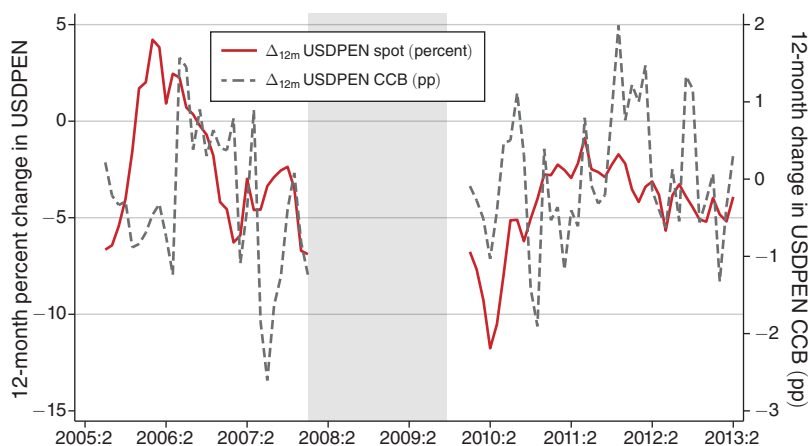


FIGURE 5. CIP DEVIATIONS AND EXCHANGE RATE

Notes: This figure plots the USDPEN cross-currency basis against the average of the cross-currency basis of other Latin American currency pairs across time. The orange line is the average of the cross-currency basis of Brazil, Chile, Colombia, and Mexico. The dotted gray line is the average of the cross-currency basis of Chile and Mexico. The red line is the cross-currency basis of Peru. Though the level of Peru's cross-currency basis is closer to the average of Brazil, Chile, Colombia, and Mexico, its movements are more correlated to those in Chile and Mexico. All of these cross-currency bases are computed using the local currency against the dollar, and they are all one-month bases. The shaded gray area represents the GFC. I am not showing these months for two reasons: to prevent an outlier period from affecting the results, and because the significant deviations affect the scale, making it hard to understand the scale in normal times.

soles, the least arbitrage-intensive banks show the greatest reduction in sol deposits as the sol depreciates. For this reason, the greater reduction in soles bank lending in banks that arbitrage more after an increase in the cross-currency basis cannot derive from the exchange rate channel.

If anything, the exchange rate channel works against finding a result in soles. Similarly, it is unlikely that the results for dollar lending are coming from the exchange rate channel, because the banks that arbitrage the most are not those with the greatest increase in dollar deposits as the basis increases and the exchange rate depreciates. I corroborate this in Table A4 in the Appendix, which shows that the baseline results are robust to adding the interaction between arbitrage intensity (β) and $\log(FX)$.

Bank Characteristics.—A concern is that bank heterogeneity explains the results. I do not find evidence that suggests that bank characteristics affect the regression results. First, the second row of Table A2 in the Appendix shows the second-stage results for a sample that includes only the four largest banks. These banks are homogeneous. The results get stronger. Second, alternative specifications that exclude all time-varying bank controls—including measures of sol and dollar deposits, total assets, profitability, and liquidity—yield results that are very similar to the baseline model. Moreover, adding bank fixed effects even strengthens the results. These results, shown in Table A.V in the online Appendix, include dropping bank controls and adding fixed effects one by one. In general, Table A.V suggests that the results are not only robust to changes in the specification regarding banks, but also to changes in the rest of the variables.

Differing Market-Based Calculations for CIP Deviations.—Another worry is that the interbank rates I use to compute the market CIP deviations do not reflect the actual cost and investment opportunities of banks. I verify the results are not driven by the exchange rate and interest rates I use to compute CIP deviations. First, Table A5 in the Appendix shows that the results are robust to using CIP deviations that account for bid-ask spreads. Second, Table A5 shows that the results are robust to using the LIBOR and the inferred put-call parity-relationship rates from van Binsberger, Diamond, and Grotteria (2021) (rows 3 and 4) instead of dollar interbank rates. Third, I compute the cross-currency basis using banks' deposit rates adjusted to include the cost of reserve requirements (row 5) and the results hold. Fourth, rows 6 and 7 show the results are robust to using alternative cross-currency basis tenors (3 months and 12 months).

These market-based calculations (rows 1–7 of Table A5) use the same security for the funding and investment legs but Section IIA describes these differ. The most important funding security for soles and dollars are bank deposits. The most important investment securities are the Central Bank of Peru's dollar deposits and soles certificates of deposit (CDs). Hence, next I calculate CIP deviations considering the direction of banks' positions and using the relevant investment and funding rates.

To do so, I consider the banking system's aggregate forward and spot position to understand whether to use sol or dollar rates for funding or investment. When the basis is negative, banks lend soles in the money market but fund soles synthetically (i.e., borrow dollars). Then, I take the dollar deposit rate as the funding rate and the soles short-term CD of the central bank as the lending rate. When the basis is positive, on the other hand, banks' positions are short dollars in forward and long in spot. Therefore, I use sol deposit rates as funding rates and dollar deposits at the central bank as investment rates. The last row of Table A5 shows the results, confirming that the simpler version of using interbank rates is a good proxy for a more rigorous computation.

Bank-Specific CIP Deviations.—The previous robustness checks use alternative specifications for market-based CIP deviations. I improve over those measures by computing bank-specific CIP deviations. These deviations rely on using the forward trade-level data, which has the initial post rate at the time the forward was closed, to obtain the forward premium for each bank. I use bank-specific interest rates on new deposits (adjusted by the reserve requirements) as funding rates. For sol investment rates, I use the rate at which each bank purchased the soles CDs of the central bank. For dollar investment rates, I use banks' dollar deposits at the central bank. The justification for using these rates comes from analyzing Table A.I in the online Appendix (see discussion in Section IIA). The CIP deviations computed with these rates are arguably the most precise measure of CIP deviations that we can obtain.

However, the analysis using bank-specific CIP deviations is subject to two important caveats. First, I cannot use an IV approach, because banks in Peru do not trade other currencies frequently. This prevents me from computing bank-specific CIP deviations for currencies besides the USDPEN pair. Because the baseline bank-lending results indicate that the OLS estimates are biased downward, we can take the results based on bank-specific CIP deviations as lower bounds. Second, the analysis should be done only with the largest banks, because smaller banks do not trade every day. Thus, we cannot compare the monthly average of bank-specific CIP

deviations across banks.²² The differences might only be a reflection of the fact that banks trade on different days. Hence, to reduce noise, I will limit the analysis to the four largest banks, but the results are very similar when using all banks.

Under these caveats, Table A.VI in the online Appendix presents the OLS estimation of the second stage of the baseline regression (equation (8b)) for Peru's largest four banks. Since I have computed the basis at the bank level, I also compute a new arbitrage intensity, $\hat{\beta}_b^{CCB_b}$, using the banks' specific CIP deviations, and I use this to interact with the bank-specific basis, CCB_b . The results are robust and of similar magnitude as the OLS estimations with the baseline regression specification. The first line shows the results using the bank-level deviations using investment and deposit rates as described. The second line, for robustness, shows the results when computing CIP deviations using adjusted deposit rates for both funding and investment. This second line shows larger effects than the baseline OLS results of online Appendix Table A.IV.

Additional Robustness Checks.—I describe additional robustness checks in online Appendix G. I show my results are robust to (i) excluding firms with foreign trade, (ii) accounting for the type of loan firms borrow (and potential selection into some banks), (iii) alternative arbitrage intensity variables, and (iv) alternative standard errors.

V. Conclusion

In this paper, I propose a channel through which CIP deviations affect bank lending. I argue that, although the existence of CIP deviations implies that banks cannot fully arbitrage CIP deviations, banks will attempt to arbitrage them when possible. To do so, banks must borrow in a particular currency. When banks cannot easily expand their balance sheets to fund the additional borrowing required to arbitrage CIP deviations, they can draw funds from their lending division and effectively decrease their lending in the currency required to perform the arbitrage. Because the arbitrage involves borrowing in a particular currency to lend in a different one, banks may substitute lending in one currency for another rather than just decreasing the total quantity lent (i.e., there is a currency change in lending, not a change of banks' balance sheet).

I document that CIP deviations can have broader effects than previously thought. They can affect firms' exchange rate exposure which is known to have important effects on the real economy after exchange rate shocks.

This result also has policy implications. As central banks in partially dollarized emerging economies try to mitigate exchange rate risk, they must understand how different policies can affect the cross-currency basis and how banks could change their lending in response to such arbitrage opportunities. Understanding this is important when analyzing whether policy interventions have the intended effect.

²²The current baseline regression specification is not affected by the problem that some banks do not trade every day. I know that if the bank does not trade forward and does not change its forward position that is offset with its spot position when the cross-currency basis changes, it is not responding to the arbitrage opportunities measured by the market. Because I can accurately measure this variable on both a daily and a monthly basis for all banks, using the arbitrage-intensity, or β_b , is comparable across banks. In fact, it accurately reflects that some banks do not trade every day by having lower betas.

For example, if central banks lean against the wind when the currency is appreciating, this could reduce the expected returns foreign investors have on carry trade investments. This is worsened if there are macro-prudential policies tackling such investments. The investments are often done by selling dollars forward. Closing such investments would generate demand for dollars forward and increase the cross-currency basis. Such an increase in the basis could, therefore, trigger the channel described in this paper. Banks would increase their dollar lending and decrease their soles lending. Effectively, they would be increasing the currency exposure of firms, which would be an unintended consequence. Taking into account the interplay of such actions and the cross-currency basis is therefore important for holistic comprehension about whether policy actions, such as exchange rate interventions, are generating the intended results.

APPENDIX

TABLE A1—SUMMARY STATISTICS OF CIP DEVIATIONS AND *FX* CHANGES

	Mean	SD	Min	Max	<i>N</i>	ρ
<i>Peru</i>						
<i>CCB</i> (%)	−0.28	0.74	−2.12	1.98	77.00	
<i>CCB</i> > 0 (%)	0.54	0.51	0.08	1.98	24.00	
<i>CCB</i> < 0 (%)	−0.65	0.49	−2.12	−0.01	53.00	
<i>CCB</i> (%)	0.62	0.50	0.01	2.12	77.00	
Δ_{1m} <i>CCB</i> (pp)	−0.04	0.68	−1.30	2.47	75.00	
Δ_{1m} <i>CCB</i> (pp)	0.51	0.45	0.01	2.47	75.00	
Δ_{12m} <i>FX</i> (%)	−3.53	3.02	−11.76	4.22	77.00	0.26
<i>Av.Latam</i>						
<i>CCB</i> (%)	−0.21	0.68	−1.91	1.41	77.00	0.43
<i>CCB</i> > 0 (%)	0.48	0.44	0.00	1.41	28.00	
<i>CCB</i> < 0 (%)	−0.61	0.42	−1.91	−0.00	49.00	
<i>CCB</i> (%)	0.56	0.43	0.00	1.91	77.00	
Δ_{1m} <i>CCB</i> (pp)	−0.01	0.39	−1.45	1.22	75.00	
Δ_{1m} <i>CCB</i> (pp)	0.28	0.27	0.00	1.45	75.00	
Δ_{12m} <i>FX</i> (%)	−5.95	6.72	−20.60	8.61	77.00	0.34
<i>Av.Chile, Mexico</i>						
<i>CCB</i> (%)	−0.03	0.55	−1.47	0.88	77.00	0.54
<i>CCB</i> > 0 (%)	0.40	0.20	0.03	0.88	43.00	
<i>CCB</i> < 0 (%)	−0.57	0.34	−1.47	−0.01	34.00	
<i>CCB</i> (%)	0.47	0.28	0.01	1.47	77.00	
Δ_{1m} <i>CCB</i> (pp)	0.00	0.28	−0.63	0.71	75.00	
Δ_{1m} <i>CCB</i> (pp)	0.22	0.17	0.01	0.71	75.00	
Δ_{12m} <i>FX</i> (%)	−4.81	6.77	−22.00	9.07	77.00	0.30

Notes: This table shows descriptive statistics of the monthly time series of the one-month cross-currency basis (*CCB*) for three groups of currencies between February 2005 and February 2013, excluding the GFC. The descriptive statistics for the USDPEN currency pair are under “Peru.” The cross-currency basis has been computed using mid-closing prices reported in Bloomberg (for *FX*) and mid-closing prices of interbank rates in dollar and soles taken from the Central Bank of Peru. The descriptive statistics of the average *CCB* of four Latin American currency pairs: Brazilian real-dollar (USDBRL), Chilean peso-dollar (USDCLP), Colombian peso-dollar (USDCOP), and Mexican peso-dollar (USDMXN) are under “Av.Latam,” while “Av.Chile, Mexico” contains those pertaining to the average cross-currency basis of USDCLP and USDMXN. I show this last group because it has greatest correlation with USDPEN. Within each group, the *CCB* (in row 1) is expressed in percentages (0–100 scale). Rows 2 and 3 describe the summary statistics narrowing the sample to periods when the basis was either positive or negative. Row 4 shows the absolute value of the *CCB*. Row 5 shows the one-month change in *CCB* in percentage points. This has not been annualized. Row 6 is similar, but uses the absolute value of *CCB*. Row 7 shows the year-over-year changes in the *FX* of that currency pair. The *CCB* rows in Av.Latam and Av.Chile, Mexico show the correlation between the *CCB* of Latam and Chile/Mexico, respectively, with the *CCB* of Peru. The *FX* row shows the correlation between the one-year *FX* changes and the one-year changes *CCB* of the corresponding countries. The last column of this table shows correlations.

TABLE A2—EFFECT OF ARBITRAGING CIP DEVIATIONS ON BANK LENDING: ALTERNATIVE SAMPLES

	Estimates					Other stats		
	log(PEN)	log(USD)	log(Total)	Ratio	log(USD)– log(PEN)	Observations	Firm CI	Month CI
1. Baseline	–25.18 (7.31)	16.16 (4.62)	3.08 (1.52)	1.45 (0.43)	41.35 (11.10)	1,348,040.00	18,374.00	77.00
2. Largest banks only	–40.87 (12.43)	37.25 (9.55)	6.74 (2.76)	3.25 (0.87)	78.12 (20.72)	1,056,886.00	16,849.00	77.00
3. Without foreign trade firms	–19.83 (6.55)	15.35 (5.42)	0.53 (1.71)	1.39 (0.48)	35.18 (10.86)	865,066.00	14,820.00	77.00
4. Commercial loans only	–22.68 (8.84)	19.46 (7.54)	1.19 (1.49)	1.76 (0.63)	42.15 (15.22)	459,178.00	10,407.00	74.00
5. Borrowing USD and PEN	–6.02 (3.30)	5.21 (2.86)	0.50 (1.57)	0.89 (0.44)	11.23 (4.66)	280,282.00	6,189.00	77.00
6. All financial institutions	–18.65 (6.16)	11.60 (3.72)	3.55 (1.52)	1.02 (0.35)	30.26 (8.93)	1,436,879.00	19,053.00	77.00
7. Extended sample Feb2005–Feb2018	–16.59 (8.43)	22.94 (10.39)	7.15 (3.35)	2.02 (0.79)	39.53 (17.09)	3,468,654.00	32,792.00	137.00

Notes: This table shows robustness checks under different samples. All specifications, unless noted otherwise, use the same fixed effects (firm-month and bank-month) and lagged bank controls as those used in the baseline specification. The dependent variables in logarithm have been multiplied by 100. The ratio of dollar loans to total loans is expressed on a 0–100 scale. The first five columns show the second-stage IV coefficients of the baseline specification (i.e., with lagged bank controls, bank-firm fixed effects, and month-firm fixed effects) under different samples. The last three columns show other statistics (number of observations, number of firm clusters, and number of month clusters). Row 1 shows the baseline second-stage regression shown in Table. Row 2 restricts the sample to Peru's largest four banks. Row 3 restricts the sample to those firms without foreign trade. Row 4 restricts the sample to commercial loans only. (To do this, I started with the loan-level dataset instead of the dataset that aggregated loans at the bank-firm-month level.) Row 5 shows the most conservative sample, limiting the analysis to only those firms borrowing in dollars and soles. Row 6 shows the coefficients for the entire financial system, adding the other financial institutions, including financials and *cajas*. These firms are subject to different banking regulations. Row 7 extends the sample out to February 2018 (the last date for which I have data). Standard errors are in parentheses. I use the dataset aggregated at the bank-firm-month level, except in row 4, where I use the dataset at the bank-firm-loan-month level.

TABLE A3—SENSITIVITY OF *FX* AND ARBITRAGE INTENSITY

	Low $\hat{\beta}$		Medium $\hat{\beta}$		High $\hat{\beta}$	
	$0 \leq \hat{\beta} < 0.2$		$1.6 \leq \hat{\beta} < 2.6$		$3.5 < \hat{\beta}$	
	Mean	SD	Mean	SD	Mean	SD
$\hat{\beta}$	0.08	0.08	2.11	0.39	4.24	0.59
Δ PEN dep/assets to 1% deprec. (pp)	–1.01	0.45	–0.33	0.21	–0.89	0.50
Δ USD dep/assets to 1% deprec. (pp)	0.53	0.24	0.83	0.05	0.62	0.76
Δ Total dep/assets to 1% deprec. (pp)	–0.47	0.56	0.49	0.16	–0.27	1.04

Notes: This table shows the summary statistics of the sensitivity of bank deposits to a 1 percent depreciation, by arbitrage intensity. The arbitrage intensity is measured by $\hat{\beta}$ and estimated using equation (7). The sensitivity of bank deposits to changes in *FX* is estimated using equation (9).

TABLE A4—BASELINE RESULTS AFTER CONTROLLING FOR *FX*

	$\log(PEN)$	$\log(USD)$	$\log(Total)$	Ratio	$\log(USD)-\log(PEN)$
$CCB_{t-1}^{Peru} \times (\hat{\beta})$	-18.06 (6.210)	15.66 (4.772)	4.886 (1.656)	1.114 (0.372)	33.71 (10.16)
$\log(FX)_{t-1} \times (\hat{\beta})$	-1.840 (0.384)	0.358 (0.303)	-0.150 (0.112)	0.0909 (0.0230)	2.198 (0.612)
Firm $\times$ Date fixed effects	Yes	Yes	Yes	Yes	Yes
Bank $\times$ Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Bank controls	Yes	Yes	Yes	Yes	Yes
Firm cluster	17,070	17,070	17,070	17,070	17,070
Month cluster	75	75	75	75	75
Observations	1,226,457	1,226,457	1,226,457	1,226,457	1,226,457

Notes: This table shows the results of the regression that modifies the baseline regression to add the interaction between $\log(FX)$ and arbitrage intensity $(\hat{\beta})$. All specifications, unless noted otherwise, use the same fixed effects (firm-month and bank-month) and lagged bank controls as those used in the baseline specification. Row 1 shows the baseline coefficients. The dependent variables in logarithm have been multiplied by 100. The ratio of dollar loans to total loans is expressed on a 0–100 scale. Standard errors are in parentheses. They result from the joint estimation with the first stage and are clustered by date and firm. The sample period is from February 2005 to February 2013, excluding the GFC. To prevent the results of the dollar loans from reflecting changes in the exchange rate, I have converted these loans to soles using a constant exchange rate (corresponding to February 2005).

TABLE A5—EFFECT OF ARBITRAGING CIP DEVIATIONS ON BANK LENDING: ALTERNATIVE COMPUTATIONS OF CIP DEVIATIONS

	<i>Estimates</i>					<i>Other stats</i>		
	$\log(PEN)$	$\log(USD)$	$\log(Total)$	Ratio	$\log(USD)-\log(PEN)$	Observations	Firm CI	Month CI
1. Baseline	-25.18 (7.31)	16.16 (4.62)	3.08 (1.52)	1.45 (0.43)	41.35 (11.10)	1,348,040.00	18,374.00	77.00
2. CCB with bid-ask spreads	-36.85 (16.83)	25.50 (12.11)	4.72 (3.42)	2.06 (0.96)	62.34 (28.09)	1,348,040.00	18,374.00	77.00
3. CCB using LIBOR rates	-46.56 (21.61)	31.93 (14.52)	6.99 (3.68)	2.76 (1.30)	78.49 (35.31)	1,348,040.00	18,374.00	77.00
4. CCB using vBDG	-44.13 (20.23)	30.96 (13.84)	7.10 (3.64)	2.62 (1.22)	75.10 (33.26)	1,348,040.00	18,374.00	77.00
5. CCB using deposit rates	-25.30 (8.15)	16.38 (5.10)	2.83 (1.51)	1.50 (0.48)	41.68 (12.53)	1,348,040.00	18,374.00	77.00
6. CCB with 3M rates	-27.58 (7.65)	20.18 (5.31)	3.73 (1.57)	1.73 (0.47)	47.76 (12.21)	1,193,597.00	18,204.00	71.00
7. CCB with 12M rates	-20.40 (5.48)	11.62 (3.27)	1.19 (1.16)	1.14 (0.30)	32.02 (7.95)	1,348,040.00	18,374.00	77.00
8. CCB Using investment and funding rates	-21.46 (8.77)	22.08 (7.34)	4.30 (1.72)	1.78 (0.63)	43.53 (15.54)	1,059,198.00	16,881.00	77.00

Notes: This table shows the second-stage results for the baseline regression when computing the alternative versions of the cross-currency basis. All specifications, unless noted otherwise, use the same fixed effects (firm-month and bank-month) and lagged bank controls as those used in the baseline specification. Row 1 shows the baseline coefficients. Row 2 shows the coefficients when computing CIP deviations using bid-ask spreads for all of these prices, for both the USDPEN basis and the USDCLP and USDMXN basis used as IV. Row 3 computes the cross-currency basis using dollar LIBOR rates instead of the dollar interbank rate. Row 4 computes the cross-currency basis using the inferred put-call parity relationship rates from van Binsberger, Diamond, and Grotteria (2021). Row 5 computes the cross-currency basis using 1-month term deposit rates, adjusted by reserve requirements. Rows 6 and 7 compute the basis using 3-month and 12-month forward rates, respectively. Row 8 computes the basis using 1-month term deposit rates, adjusted by reserve requirements, as funding rates in *USD* and *PEN*, short-term central bank soles CD rates for *PEN* investment rates, and overnight dollar deposit at the central bank rates for *USD* investment rates. The dependent variables in logarithm have been multiplied by 100. The ratio of dollar loans to total loans is expressed on a 0–100 scale. Standard errors are in parentheses. They result from the joint estimation with the first stage and are clustered by date and firm. The sample period is from February 2005 to February 2013, excluding the GFC. To prevent the results of the dollar loans from reflecting changes in the exchange rate, I have converted these loans to soles using a constant exchange rate (corresponding to February 2005).

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